

ONLINE FOOD DELIVERY BUSINESS INTELLIGENCE & PERFORMANCE ANALYSIS USING SQL

PROJECT REPORT

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ABSTRACT

The Online Food Delivery Business Intelligence & Performance Analysis project focuses on analyzing transactional and operational data from an online food delivery platform using SQL. The database consists of customers, restaurants, orders, order items, and delivery agents. The objective of the project is to generate meaningful business insights related to revenue performance, customer retention, restaurant growth, product popularity, and operational efficiency.

Using SQL queries, various Key Performance Indicators (KPIs) such as total revenue, customer growth, average order value, top-performing restaurants, and delivery performance are analyzed. The findings help businesses make data-driven decisions, improve customer satisfaction, optimize delivery operations, and increase profitability.

1. INTRODUCTION

The Online Food Delivery Business Intelligence & Performance Analysis project aims to analyze business data generated from an online food delivery platform. The project utilizes a relational database containing customer, restaurant, order, product, and delivery information to understand business performance and operational trends.

With the rapid growth of online food delivery services, organizations need data-driven insights to improve customer retention, maximize revenue, and enhance operational efficiency. SQL serves as the primary analytical tool for extracting, transforming, and analyzing data to support strategic decision-making.

This project simulates real-world business intelligence practices used by modern food delivery companies and demonstrates practical SQL skills for business analytics.

2. PROBLEM STATEMENT

Online food delivery companies generate large volumes of transactional data every day. However, without proper analysis, businesses may struggle to understand customer preferences, identify high-performing restaurants, evaluate delivery efficiency, and monitor revenue growth.

The project aims to answer critical business questions such as:

- What is the total revenue generated?
- Which restaurants contribute the most revenue?
- Which products are most popular?
- Which cities generate the highest sales?
- How effective are discounts in driving sales?
- How efficient is the delivery process?

The answers to these questions help management make informed decisions and improve overall business performance.

3. BUSINESS OBJECTIVES

- Analyze revenue performance and sales trends.
 - Improve customer retention through customer behavior analysis.
 - Evaluate restaurant performance and growth.
 - Identify best-selling products and cuisines.
 - Measure delivery efficiency and operational performance.
 - Analyze payment method preferences.
 - Compare city-wise business performance.
 - Generate business KPIs for strategic decision-making.
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4. DATASET OVERVIEW

Dataset Description

The dataset represents an online food delivery platform and contains customer, restaurant, order, item, and delivery-related information.

Tables Used

Customers

- customer_id
- name
- email
- city
- signup_date
- gender

Restaurants

- restaurant_id
- rest_name
- city
- cuisine
- rating

Orders

- order_id
- customer_id
- restaurant_id
- order_date
- order_amount
- discount
- payment_method
- delivery_time

Order_Item

- order_item_id
- order_id
- item_name
- quantity
- price

Delivery_Agents

- agent_id
 - agent_name
 - city
 - joining_date
 - rating
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5. ENTITY RELATIONSHIP (ER) DIAGRAM

Entities

- Customers
- Restaurants
- Orders
- Order_Items
- Delivery_Agents

Relationships

- One Customer → Many Orders
- One Restaurant → Many Orders
- One Order → Many Order Items
- One Delivery Agent → Many Orders

Primary Keys

- customer_id
- restaurant_id
- order_id
- order_item_id
- agent_id

Foreign Keys

- customer_id → Customers
 - restaurant_id → Restaurants
 - order_id → Order_Items
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6. DATABASE DESIGN

The database follows a relational structure where each table stores specific information while maintaining relationships through primary and foreign keys. This design minimizes redundancy and ensures data integrity.

7. SQL ANALYSIS & BUSINESS QUESTIONS

Query 1: Total Revenue

SQL Query

```
sql id="lyys5q" SELECT SUM(order_amount - discount) AS total_revenue FROM orders;
```

Business Question

What is the total revenue generated after discounts?

Insight

Provides net revenue generated by the platform.

Query 2: Total Orders

SQL Query

```
sql id="4t0ldf" SELECT COUNT(*) AS total_orders FROM orders;
```

Business Question

How many orders were placed?

Insight

Measures overall business activity.

Query 3: Average Order Value

SQL Query

```
sql id="x57y17" SELECT AVG(order_amount) AS avg_order_value FROM orders;
```

Business Question

What is the average spending per order?

Insight

Helps evaluate customer purchasing behavior.

Query 4: Top Performing Restaurants

SQL Query

```
sql id="n98o92" SELECT r.rest_name, SUM(o.order_amount - o.discount)
AS revenue FROM orders o JOIN restaurants r ON o.restaurant_id =
r.restaurant_id GROUP BY r.rest_name ORDER BY revenue DESC;
```

Insight

Identifies restaurants generating the highest revenue.

Query 5: Most Popular Food Items

SQL Query

```
sql id="0vc08q" SELECT item_name, SUM(quantity) AS total_sold FROM
order_item GROUP BY item_name ORDER BY total_sold DESC;
```

Insight

Shows best-selling food items.

Query 6: Payment Method Analysis

SQL Query

```
sql id="515q0h" SELECT payment_method, COUNT(*) AS total_orders FROM
orders GROUP BY payment_method;
```

Insight

Identifies customer payment preferences.

Query 7: City-wise Revenue Analysis

SQL Query

```
sql id="womdx4" SELECT c.city, SUM(o.order_amount - o.discount) AS  
revenue FROM orders o JOIN customers c ON o.customer_id = c.customer_id GROUP  
BY c.city;
```

Insight

Determines top revenue-generating cities.

Query 8: Delivery Performance

SQL Query

```
sql id="5w3i9m" SELECT AVG(delivery_time) AS avg_delivery_time FROM orders;
```

Insight

Measures operational efficiency.

8. KEY PERFORMANCE INDICATORS (KPIs)

- Total Revenue
 - Total Orders
 - Total Customers
 - Average Order Value
 - Customer Retention Rate
 - Average Delivery Time
 - Top Restaurant Revenue
 - Most Popular Food Item
 - City-wise Revenue
 - Payment Method Distribution
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9. BUSINESS INSIGHTS

- Certain restaurants contribute significantly to overall revenue.
- Popular food items drive repeat purchases.
- Faster delivery improves customer satisfaction.

- Digital payment methods dominate transactions.
 - Revenue varies significantly across cities.
 - Discounts influence customer ordering behavior.
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10. CONCLUSION

The Online Food Delivery Business Intelligence & Performance Analysis project successfully demonstrates the use of SQL for analyzing real-world business data. Through customer, restaurant, sales, and delivery analysis, valuable insights are generated that support data-driven decision-making.

The project highlights how SQL can be used to transform raw data into actionable business intelligence, helping organizations improve profitability, customer satisfaction, and operational efficiency.

11. FUTURE ENHANCEMENTS

- Build interactive dashboards using Power BI or Tableau.
 - Implement customer segmentation analysis.
 - Develop predictive models for sales forecasting.
 - Analyze customer lifetime value (CLV).
 - Integrate real-time analytics.
 - Create automated KPI reporting systems.
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12. REFERENCES

- SQL Documentation
- MySQL Documentation
- Database Management System Concepts
- Business Intelligence and Data Analytics Resources
- Online Food Delivery Industry Reports